

CSA0614-TOPIC 4-DAA

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Ask Gemini

SIMATS ENGINEERING SIMATS Online Programming Lab DAA PROGRAMMING YANAMALA GREESHMA REDDY

LAB PROGRAM- TOPIC 4

Language Python C++ Java

QUESTIONS

Q1 Closest Pair of Points (Brute Force Approach) 10 marks

Q2 Closest Pair of Points (Brute Force Approach) 10 marks

Q3 Closest Pair of Points (Brute Force Approach) 10 marks

Q4 Closest Pair of Points (Brute Force Approach) 10 marks

Q5 Closest Pair of Points (Brute Force Approach) 10 marks

Q6 Closest Pair of Points (Brute Force Approach) 10 marks

Q7 Closest Pair of Points (Brute Force Approach) 10 marks

Q8 Closest Pair of Points (Brute Force Approach) 10 marks

Q9 Closest Pair of Points (Brute Force Approach) 10 marks

Q10 Closest Pair of Points (Brute Force Approach) 10 marks

10/10 answered

Closest Pair of Points (Brute Force Approach) 10 marks

Write a program that finds the closest pair of points in a set of 2D points using the brute force approach.

Input:

- A list of points represented by coordinates (x, y).
- Points (1, 2), (4, 5), (7, 8), (3, 10).

Output:

- The two points with the minimum distance.
- The minimum distance between them.

Example Output:

Closest pair: (1, 2) - (3, 10)
Minimum distance: 1.4142135623730951

Your Code

```
1 import math
2
3 def calculate_distance(p1, p2):
4     return math.sqrt((p1[0] - p2[0])**2 + (p1[1] - p2[1])**2)
5
6 def find_closest_pair(points):
7     min_dist = float('inf')
8     closest_pair = None
9     n = len(points)
10
11     for i in range(n):
12         for j in range(i+1, n):
13             dist = calculate_distance(points[i], points[j])
14             if dist < min_dist:
15                 min_dist = dist
16                 closest_pair = (points[i], points[j])
17
18     return closest_pair, min_dist
19
20 if __name__ == "__main__":
21     points = [(1, 2), (4, 5), (7, 8), (3, 10)]
22     pair, distance = find_closest_pair(points)
23     print(f"Closest pair: {pair[0]} - {pair[1]}")
24     print(f"Minimum distance: {distance}")
```

Input

(1,2)-(3,10)

Output

Closest pair: (1, 2) - (3, 10)
Minimum distance: 1.4142135623730951

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Q10 Closest Pair of Points (Brute Force Approach) 10 marks

10/10 answered

Closest Pair of Points (Brute Force Approach) 10 marks

Develop a program to calculate the shortest distance between any two points using the brute force closest pair algorithm.

Input:

Points (3, 2), (2, 3), (5, 5)

Output:

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

Your Code

```
1 import math
2
3 def calculate_distance(p1, p2):
4     return math.sqrt((p1[0] - p2[0])**2 + (p1[1] - p2[1])**2)
5
6 def find_closest_pair(points):
7     min_dist = float('inf')
8     closest_pair = None
9     n = len(points)
10
11     for i in range(n):
12         for j in range(i+1, n):
13             dist = calculate_distance(points[i], points[j])
14             if dist < min_dist:
15                 min_dist = dist
16                 closest_pair = (points[i], points[j])
17
18     return closest_pair, min_dist
19
20 if __name__ == "__main__":
21     points = [(3, 2), (2, 3), (5, 5)]
22     pair, distance = find_closest_pair(points)
23     print(f"Closest pair: {pair[0]} - {pair[1]}")
24     print(f"Minimum distance: {distance}")
```

Input

(3,2)-(2,3)-(5,5)

Output

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

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